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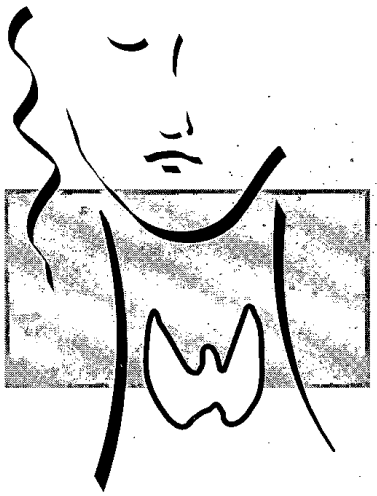
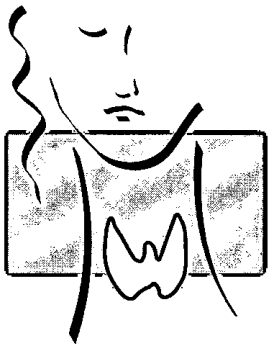
Understanding Your Child's Thyroid



Why Is the Thyroid Gland So Important in Childhood?

In order to understand the importance of the thyroid gland — and the possibilities of thyroid disease in your child — you first need to understand the normal thyroid gland and what it does.



If You Need More Help...

For more information on thyroid disease, contact:

The American Thyroid Association
Montefiore Medical Center
111 E. 210th St.
New York, NY 10467
800–542–6687

The Thyroid Foundation of America
Ruth Sleeper Hall, RSL 350
40 Parkman St.
Boston, MA 02114–2698
(send self-addressed, stamped envelope)

The Thyroid Society for Education & Research
7515 S. Main St., Suite 545
Houston, TX 77030
800-THYROID

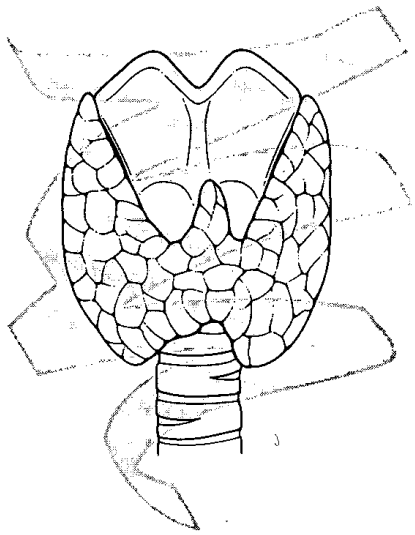
National Graves' Disease Foundation
320 Arlington Rd., Jacksonville, FL 32211

3) *Surgery* — Most of the thyroid gland is removed surgically to stop hormone production. In most cases, thyroid hormone therapy is then needed to replace the normal thyroid hormone.

What You Should Know if Your Child has to Take Thyroid Hormone Medication

If your child is placed on thyroid hormone therapy, you should know that all thyroid hormone preparations are not the same. If the doctor prescribes a certain brand of levothyroxine, a synthetic thyroid hormone, and your pharmacist asks if you would like to switch brands, ask the doctor first. Most thyroid experts agree that when thyroid medications are switched the patient should be retested and the dose should be adjusted if necessary.

The Thyroid Gland



The thyroid, a small gland in the front of the neck just below the Adam's apple, produces and secretes thyroid hormone. It is shaped like a butterfly, with one wing on either side of the windpipe. If the thyroid gland becomes diseased or injured, it may not work properly.

People are said to be *hypo*thyroid when the thyroid gland cannot produce enough thyroid hormone. "Hypo" means "too little." People whose thyroid glands are too active are said to be *hyper*thyroid. "Hyper" means "too much" — and it means that the thyroid gland is producing too much hormone.

What the Thyroid Gland Does

The thyroid produces hormones that collectively are called thyroid hormone. The hormones are carried by the bloodstream to every cell in the body. Within the cells, these hormones control metabolism — the rate at which the body produces energy from nutrients. The growth of cells and the working of muscles depend on having the correct level of thyroid hormone. Your child's temperature, cholesterol level, moods, memory, and many other things are all affected by this hormone.

The heart, brain, liver, kidneys, and skin all need the right amount of thyroid hormone to work effectively. Thyroid hormone also affects muscle strength, reproductive functions, and appetite. Problems arise when this gland does not function properly and begins producing thyroid hormone in the wrong amounts. This is referred to as thyroid disorder. Too little or too much thyroid hormone can create a wide range of signs and symptoms. Thyroid disorders (both "hyper" and "hypo") affect more than 11 million Americans.

function should be tested frequently during growth years and — depending on the results — the dosage may be adjusted.

Treatment of an Overactive Thyroid Gland

If the goiter has caused an overactive thyroid gland, there are three treatment alternatives:

1) *Drug treatment* — These drugs prevent the thyroid from manufacturing hormone so that symptoms gradually stop — usually within 6 to 8 weeks.

2) *Radioactive iodine treatment* — This substance goes into the thyroid gland and inactivates thyroid tissue so that it cannot produce thyroid hormone any longer. With this method, symptoms are usually gone within 3 to 6 months. In most cases, thyroid hormone therapy is then needed to replace the normal thyroid hormone.

The TSH test is considered a major advance in testing. Most thyroid experts feel that the TSH test is the frontline test for checking thyroid function. A sample of your child's blood will be analyzed with the TSH test and can usually determine the presence of thyroid disease.

Another blood test that can be used to diagnose thyroid disease is the T_4 (thyroxine) test. If your child's T_4 level is abnormal, this may confirm the presence of thyroid disease. If the diagnosis is still uncertain, there are other tests that can be easily performed in a doctor's office to confirm the diagnosis.

Treatment of an Underactive Thyroid Gland

If the thyroid disease has caused an underactive thyroid gland, it is simple to treat with thyroid hormone therapy. This medication replaces the body's natural hormone. Once therapy starts, it is important for your child to continue seeing the doctor for regular checkups. His or her thyroid

It is important that adults (parents, teachers, and pediatricians) be on the alert for thyroid problems in children. Parents in particular should be aware of a family history of thyroid problems since some thyroid disorders are inherited.

Thyroid Problems in Children can Vary

When the thyroid gland does not work properly, children may display different signs and symptoms than those of an adult. This is because a child is growing — and growth and development can be influenced by an overactive or underactive thyroid gland.

It may be harder to recognize a thyroid problem in a child than in an adult. Since children don't know what "normal" is, they are less likely to complain about feeling sick. Even older children may simply accept the emotional and physical symptoms of a thyroid problem as "normal" for them.

A more obvious sign of thyroid disease occurs when the thyroid gland grows larger than normal. This is called a *goiter*. A goiter may occur at any age but it is especially common in girls around the time of puberty.

It is possible that there may not be anything different except an enlargement of the throat. If your child *does* have symptoms, however, how will they feel? What should you look for?

What to look for will depend on whether the thyroid gland is over- or underactive. Either one of these conditions will interfere with your child's growth and development.

SIGNS AND SYMPTOMS OF AN OVERACTIVE THYROID GLAND

- Nervousness
- Irritability
- Difficulty sleeping
- Bulging eyes
- Unblinking stare
- Goiter
- Rapid heartbeat
- Increased sweating
- Heat intolerance
- Unexplained weight loss
- Increased appetite
- Frequent bowel movements
- Warm, moist palms
- Fine tremor of fingers

Congenital Hypothyroidism

Underactive Thyroid Gland in an Infant

In the United States, one in every 4000 to 5000 babies is born with hypothyroidism. If a baby is born with an underactive thyroid, he or she may seem normal at birth. Often these

Diagnosing Thyroid Disease

If your doctor suspects thyroid disease in your child, the first step will be to perform a physical examination to look for evidence of enlargement of the gland. Next, a blood sample will be drawn and, more than likely, a sensitive TSH test will be done (TSH stands for Thyroid Stimulating Hormone).

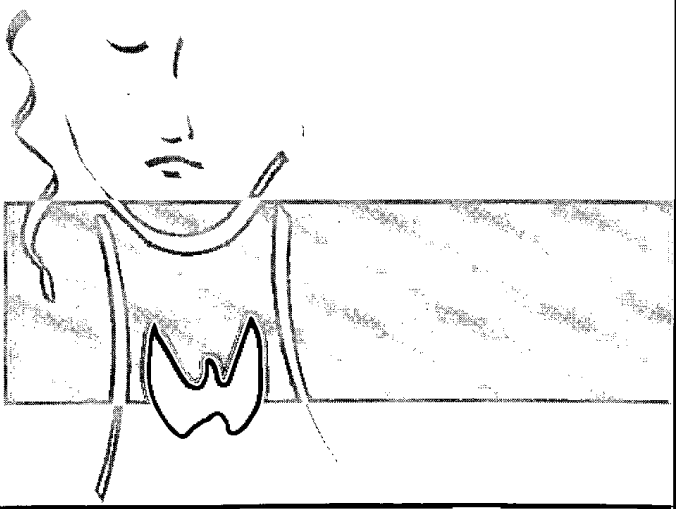
An Overactive Thyroid Gland in an Older Child

With an overactive thyroid gland, the symptoms in an older child are not very different from those of an adult. You may feel your child's heart pounding or beating too fast — even when he or she is at rest. You may notice your child acting "wired," restless, and/or unable to sleep.

Sometimes the only clue will be a sudden growth spurt — which parents will notice when a child starts outgrowing his or her clothes very rapidly. Parents and teachers may also note poor school performance and lack of attention. Shaky hands may cause clumsiness and poor handwriting, while weak shoulder and thigh muscles may be noticed in sports or at play. You may also notice one or more of the following symptoms:

babies will appear somewhat jaundiced, but this will slowly clear. In the following days and weeks the infant may become sluggish, sleepy, and develop a hoarse cry, constipation, stuffy nose, and puffy face typical of a hypothyroid condition known as *cretinism*. But by this time, the baby is often at home and it may be difficult to realize that he or she is seriously sick.

If these babies are not treated as soon as possible after birth, permanent brain damage and mental retardation can occur. Fortunately, if the condition is discovered and treated early, permanent damage can usually be avoided. However, it is extremely important for this condition to be diagnosed as early as possible, preferably with blood tests done at birth. Newborn screening for congenital hypothyroidism is required in all 50 states.



Acquired Hypothyroidism

Underactive Thyroid Gland in an Older Child

If your child's thyroid becomes underactive (sometimes caused by chronic lymphocytic thyroiditis), you may notice your child acting sluggish and tired — even when they've had enough sleep. One of the first signs of hypothyroidism may be a swollen thyroid gland and noticeable slowing of their growth pattern. You may also notice one or more of the following symptoms:

SIGNS AND SYMPTOMS OF AN UNDERACTIVE THYROID GLAND

- Growth retardation
- Weakness
- Forgetfulness
- Dry, coarse hair
- Puffy face and eyes
- Goiter
- Slow heartbeat
- Dry skin
- Cold intolerance
- Weight gain
- Constipation

An Overactive Thyroid Gland in an Infant

One of the rarest often-studied thyroid conditions in children is called *neonatal hyperthyroidism.* This condition can occur in a baby whose mother is hyperthyroid during pregnancy. Signs and symptoms of hyperthyroidism in infants include goiter, premature birth, fever, flushing, restlessness, and rapid heartbeat. If the disease appears in its mildest form, no treatment may be needed as the disease may subside on its own. If the baby is seriously ill, however, treatment with iodine and an antithyroid drug may be required. Even though this condition is rare, it is important for a pregnant woman with a history of hyperthyroidism to alert her obstetrician about her problem during pregnancy.